

Naive Lie Theory 3.5 Exercises

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Notes

I am going to use this in 3.5.3: A surjective homomorphism $\varphi : G \rightarrow H$ maps center into center. Here is the proof: say $g \in Z(G)$, so $gg' = g'g$ for all $g' \in G$. The homomorphism gives $\varphi(g)\varphi(g') = \varphi(g')\varphi(g)$. Because φ is surjective, so for all $h' \in H$ there's a g' such that $\varphi(g') = h'$. Therefore $\varphi(g)h' = h'\varphi(g)$ for all $h' \in H$ so $\varphi(g) \in Z(H)$.

3.5.1

The subgroup made by matrices of the form

$$R_\theta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & -e^{i\theta} \end{pmatrix}$$

3.5.2

$\mathrm{SO}(3)$ is isomorphic to the quotient group $\mathrm{SU}(2)/\{I, -I\}$. Recall that the maximal torus of $\mathrm{SO}(3)$ is $\mathbb{T}^1$. Obviously $\mathrm{SO}(3) = \mathrm{SU}(2)/\{I, -I\}$ is a subgroup of $\mathrm{SU}(2)$, so the maximal torus of $\mathrm{SU}(2)$ is at least $\mathbb{T}^1$.

For the sake of contradiction, assume $H = \mathbb{T}^2$ is a subgroup of $\mathrm{SU}(2)$. Say H is a direct product of two $\mathbb{T}^1$, A and B . WLOG consider the following two cases:

1. If $-I \notin A$, the homomorphism Φ restricted to A is an isomorphism, so its image in $\mathrm{SO}(3)$ is still a circle.
2. If $-I \in A$, $\Phi(A)$ is the quotient group $A/\{I, -I\}$, i.e. a circle with antipodal points identified by each other, which is still a circle. As an example, consider the unit circle in $\mathbb{C}$, if one identifies z with $-z$, equivalently, one is giving $z \rightarrow z^2$, which is still the same circle.

Similar arguments can be made on B . If $\Phi(A)$ and $\Phi(B)$ are the same circle, the whole $\mathbb{T}^2$ is 2-to-1 mapped to a circle, which is impossible. So, they are distinct and form a $\mathbb{T}^2$. Contradiction.

3.5.3

Using the lemma mentioned in Notes, the image of $Z(\mathrm{SU}(2))$ must be in $Z(\mathrm{SO}(3))$. Recall that $Z(\mathrm{SO}(3)) = \{I\}$, so $Z(\mathrm{SU}(2))$ is just the preimage of $\{I\}$, which is $\{I, -I\}$.

3.5.4

Try it on a rectangular object but not a cube, like a phone, book, etc.

3.5.5

A $\frac{\pi}{2}$ rotation about e_3 axis.

3.5.6

$-I$.